

Manuscripts considered — hgsoc-tuboovarian-stage3c-r0-hrd-pen ding-idyq

Master inventory: every paper reviewed in this case

Generated 2026-06-26

Manuscripts considered — hgsoc-tuboovarian-stage3c-r0-hrd-per

Master inventory: 8 clinical (8 included, 0 considered & excluded) + 11 pre-clinical (9 included, 2 considered & excluded) + 8 additional trial publications/registrations. One entry per manuscript, sorted by year (newest first). Excluded entries are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

—/— (2025) — Sci Rep

Reference:doi:10.1038/s41598-025-90714-9 · **Type:** preclinical · **Inclusion:** excluded — PI3Kalpha inhibitor (BYL-719/alpelisib) plus cisplatin synergy was shown in PIK3CA-mutant ovarian cells in vitro only, with no in-vivo arm, and keys on PIK3CA hotspot mutation rather than the patient's amplification. The PI3K/HRD axis is already covered by the Wang 2016 in-vivo row; logged for audit trail.

- **Intervention:** PI3Kalpha inhibitor BYL-719 plus cisplatin in PIK3CA-mutant ovarian cells
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: PI3Kalpha inhibitor (BYL-719/alpelisib) plus cisplatin synergy was shown in PIK3CA-mutant ovarian cells in vitro only, with no in-vivo arm, and keys on PIK3CA hotspot mutation rather than the patient's amplification. The PI3K/HRD axis is already covered by the Wang 2016 in-vivo row; logged for audit trail.

Thaker/Anwer (2024) — Gynecol Oncol (2025) / sponsor topline

Reference: — · **Type:** clinical · **Inclusion:** included

- **Intervention:** GEN-1 / IMNN-001 (IL-12 immunogene)
- **Indication / model:** newly diagnosed advanced epithelial ovarian cancer, neoadjuvant + interval debulking (1L neoadjuvant); Advanced epithelial ovarian cancer treated with neoadjuvant chemotherapy plus IP IL-12 immunogene. Regimen-matched to this patient, who received GEN-1 with neoadjuvant carbo/paclitaxel and achieved R0. Acts via the tumor immune microenvironment, so the foreclosed MSS/TMB-low checkpoint axis does not exclude it.
- **Design:** randomized phase 1/2 added to neoadjuvant chemotherapy (OVATION 2 / NCT03393884)
- **n:** 110
- **Effect size:** 0.79 HR — median PFS (ITT)
- **Toxicities:** Sponsor and abstract report IP delivery generally well tolerated with no new safety signal; no per-term grade 3+ AE table in the public topline / abstract. No named patient toxicity veto implicated.
- **Case fit:** strong
- **Notes:** Regimen-matched investigational program the patient is already on. Efficacy from sponsor topline (Imunon, Oct 2024) and the Gynecol Oncol 2025 abstract; the full peer-reviewed paper (publisher page 403) and ClinicalTrials.gov AE tab were not accessible, so no per-term toxicities extracted - backfill when full text opens. CI bounds not reported in the topline. FDA has cleared a phase 3 program.

—/— (2024)

Reference: — · **Type:** preclinical · **Inclusion:** excluded — No primary preclinical study ties a MAP2K4 (MKK4) deletion to a tractable, validated drug target in ovarian cancer; the literature treats MAP2K4 loss as a poorly-characterized JNK/p38-pathway event with no matched agent. Nothing meets the bar for an included mechanism row, so this records the negative search result rather than logging trial-list baskets as preclinical

evidence.

- **Intervention:** MAP2K4 (MKK4) deletion as a drug target in ovarian cancer
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: No primary preclinical study ties a MAP2K4 (MKK4) deletion to a tractable, validated drug target in ovarian cancer; the literature treats MAP2K4 loss as a poorly-characterized JNK/p38-pathway event with no matched agent. Nothing meets the bar for an included mechanism row, so this records the negative search result rather than logging trial-list baskets as preclinical evidence.

Thaker/Anwer (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** GEN-1 (IMNN-001) IL-12 immunogene + neoadjuvant carboplatin/paclitaxel
- **Indication / model:** newly-diagnosed advanced epithelial ovarian cancer (neoadjuvant + interval debulking) (1L neoadjuvant); none (IL-12 immunogene, tumor immune microenvironment)
- **Design:** randomized phase 1/2 added to neoadjuvant carboplatin/paclitaxel (OVATION 2)
- **n:** 110
- **Effect size:** — — progression-free survival / R0 resection rate
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** PI3K/AKT-pathway inhibitor
- **Indication / model:** advanced solid tumors with PIK3CA / PI3K-pathway alterations (basket) (2L+); PIK3CA alteration (mutation or amplification)
- **Design:** phase 1 biomarker-selected basket
- **Effect size:** — — safety / pharmacodynamics / ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** investigational agent matched to MAP2K4 loss
- **Indication / model:** advanced solid tumors (biomarker-matched basket including MAP2K4 loss) (2L+); MAP2K4 deletion / loss-of-function
- **Design:** phase 1 biomarker-matched basket
- **Effect size:** — — safety / ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Monk/Monk (2022) — J Clin Oncol

Reference:PMID 35658487 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Rucaparib (maintenance)
- **Indication / model:** newly diagnosed advanced high-grade epithelial ovarian cancer, response after 1L platinum doublet (maintenance); All-comers (4:1 randomization), HRD-positive and HRD-negative strata. Benefit held in the ITT population and in the non-nested HRD-negative subgroup, so a BRCA-WT, HRD-unquantified tumor still qualifies.
- **Design:** randomized double-blind placebo-controlled phase 3, 1L maintenance (ATHENA-MONO / GOG-3020 / ENGOT-ov45)
- **n:** 538
- **Effect size:** 0.52 HR — median PFS (intent-to-treat)
- **Variance:** 95% CI 0.4–0.68; $p < 0.0001$
- **Toxicities:** anemia $G \geq 3$ (28.7%); neutropenia $G \geq 3$ (14.6%); ALT/AST elevation $G \geq 3$ (10.6%); treatment discontinuation (AE) (11.8%)
- **Case fit:** strong
- **Notes:** Second all-comer 1L PARP-maintenance option. ITT HR 0.52 (95% CI 0.40-0.68) from the primary JCO publication (pmid 35658487). Qualifies the BRCA-WT tumor via the ITT/HRD-negative benefit, not a BRCA gate. Severe-myelosuppression veto applies.

—/— (2022)

Reference: — · **Type:** clinical · **Inclusion:** included

- **Intervention:** PI3K/AKT-pathway inhibitor (investigational, PIK3CA-directed)
- **Indication / model:** advanced solid tumors with PIK3CA alterations (biomarker-selected basket) (2L+); Biomarker-matched baskets keyed on PIK3CA mutation OR amplification. The patient has focal somatic PIK3CA amplification; benefit data for amplification (as opposed to activating hotspot mutation) are weak, and there is no approved PI3K-directed therapy in ovarian cancer.
- **Design:** phase 1 dose-escalation/expansion baskets (e.g. NCT05216432, NCT05683418)
- **Effect size:** — — — objective response rate / safety (no mature efficacy for PIK3CA-amplified ovarian cancer)
- **Toxicities:** PI3Kalpha-pathway class toxicity is hyperglycemia, rash, diarrhea, and stomatitis; no published per-term rates for the specific PIK3CA-amplified ovarian-cancer context. No named patient veto directly implicated.
- **Case fit:** weak
- **Notes:** Context-only row kept clearly investigational. There is no approved PI3K/AKT therapy in HGSOC and amplification is a poorly-validated predictive marker versus activating mutation. Surfaced so the board sees the PI3K axis the Altera report flagged; not a ranked option. No primary-source efficacy or AE data to extract (early-phase baskets without ovarian-specific readouts).

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** PI3Kalpha-selective inhibitor (mutant/amplification-directed)
- **Indication / model:** advanced solid tumors with PIK3CA alterations (basket) (2L+); PIK3CA mutation or amplification (PI3Kalpha-pathway activation)
- **Design:** phase 1 dose-escalation/expansion basket, biomarker-selected
- **Effect size:** — — safety / objective response rate

- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** investigational agent matched to MAP2K4 loss
- **Indication / model:** advanced solid tumors (biomarker-matched basket including MAP2K4 loss) (2L+); MAP2K4 deletion / loss-of-function
- **Design:** phase 1 biomarker-matched basket
- **Effect size:** — — safety / ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Gonzalez-Martin/Mirza (2019) — N Engl J Med

Reference: [PMID 31562799](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Niraparib (maintenance)
- **Indication / model:** newly diagnosed advanced high-grade serous/endometrioid ovarian cancer, response after 1L platinum (maintenance); All-comers, both HRD-positive and HRD-negative strata; no BRCA mutation and no validated HRD score required for enrolment. The all-comer enrolment is what makes PRIMA the cleanest fit for this BRCA-proficient, genomically-unstable tumor.
- **Design:** randomized double-blind placebo-controlled phase 3 (2:1), 1L maintenance (PRIMA/ENGOT-OV26)
- **n:** 733
- **Effect size:** 0.62 HR — median PFS (overall population)
- **Variance:** 95% CI 0.5–0.76; $p < 0.001$
- **Toxicities:** anemia $G \geq 3$ (31%); thrombocytopenia $G \geq 3$ (28.7%); neutropenia $G \geq 3$ (12.8%)
- **Case fit:** strong
- **Notes:** Pivotal 1L all-comer maintenance trial behind niraparib's broad label. The BRCA-WT, genomically-unstable profile here qualifies via the all-comer rationale rather than any BRCA gate; HRD-positive subgroup shows larger benefit but enrolment never required it. Triggers the patient's severe-myelosuppression toxicity veto, partly mitigated by individualized starting dose.

Ray-Coquard/Pujade-Lauraine (2019) — N Engl J Med

Reference: [PMID 31851799](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Olaparib + bevacizumab (maintenance)
- **Indication / model:** newly diagnosed advanced high-grade serous ovarian cancer receiving 1L bevacizumab (maintenance); All-comers enrolled, but the predefined and label-relevant benefit is in the HRD-positive subgroup (Myriad MyChoice GIS ≥ 42 or tumor BRCA mutation). This patient is BRCA wild-type and has only a qualitative Altera instability call, so eligibility is CONTINGENT on a validated GIS she does not yet have.
- **Design:** randomized double-blind placebo-controlled phase 3, olaparib vs placebo added to bevacizumab backbone (PAOLA-1 / ENGOT-ov25)
- **n:** 806
- **Effect size:** 0.33 HR — median PFS (HRD-positive subgroup)

- **Variance:** 95% CI 0.25–0.45; $p < 0.001$
- **Toxicities:** hypertension $G \geq 3$ (19%); anemia $G \geq 3$ (17%); lymphopenia $G \geq 3$ (7%); fatigue/asthenia $G \geq 3$ (5%)
- **Case fit:** partial
- **Notes:** Highest-magnitude PARP option (HRD-positive HR 0.33) and the patient is already on the bevacizumab backbone, but it is contingent: PAOLA-1 benefit concentrates in HRD-positive disease defined by MyChoice GIS ≥ 42 or tumor BRCA. Altera's qualitative instability call does not meet that companion-diagnostic threshold. The BRCA-non-mutated HRD-positive subgroup (HR 0.43) is the relevant comparator only if a validated GIS comes back ≥ 42 ; below that she is HRD-negative and defaults to all-comer niraparib. Combination-arm grade 3+ AE per PMC10859514 expert review.

Kaplan/Glazer (2019) — Sci Transl Med

Reference: [PMID 31092693](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Cediranib induces a homologous-recombination defect that sensitizes BRCA-wild-type tumors to olaparib
- **Indication / model:** BRCA-wild-type ovarian lines (IGROV1, SKOV3, PEO14) plus BRCA2-mutant PEO1/PEO4 and breast lines; IGROV1 and hypoxia-reporter xenografts in nu/nu mice
- **Design:** Cediranib drives down BRCA1, BRCA2, and RAD51 through two routes: VEGFR2-mediated tumor hypoxia, and a hypoxia-independent PDGFR-blockade pathway that activates PP2A and E2F4/p130 transcriptional repression. The result is a contextual, drug-induced HR defect in cells that are genetically HR-proficient.
- **n:** multi-line panel; xenograft arms (group sizes per arm not specified in text)
- **Effect size:** strong — Cediranib produced an HR-deficient state in BRCA-wild-type ovarian cells and sensitized them to olaparib both in culture and in xenografts. This is the mechanistic basis for combining a VEGFR inhibitor with a PARP inhibitor in BRCA-proficient HGSC, the exact genetic context of this tumor.
- **Variance:** vs vehicle; single-agent olaparib; single-agent cediranib; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Cediranib is the tool VEGFR-TKI here, not the patient's bevacizumab; the induced-HRD concept transfers to anti-angiogenic plus PARP strategy but the specific agent differs. Subcutaneous and reporter xenografts rather than orthotopic peritoneal disease.

Heinzelmann-Schwarz/Jacob (2018) — Gynecol Oncol

Reference: [PMID 29157627](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Letrozole (endocrine maintenance)
- **Indication / model:** ER-positive high-grade serous ovarian cancer, maintenance / later-line (maintenance); ER-positive HGSOC by IHC; a translational study with a clinical letrozole-maintenance series, enriched for platinum-resistant and residual-disease patients. The patient is ER-positive (~80% nuclei) but PR-negative, which tempers expected magnitude.
- **Design:** translational study with retrospective clinical letrozole-maintenance comparison (TCGA/IHC analysis + treated cohort)
- **Effect size:** 60% vs 38.5% RFS at 24 mo % — recurrence-free interval at 24 months
- **Variance:** $p = 0.035$
- **Toxicities:** No per-term toxicity table in the source; aromatase-inhibitor class toxicity is arthralgia and

bone-density loss on prolonged use. Oral, low-burden, no named patient veto applies.

- **Case fit:** partial
- **Notes:** Low-toxicity oral option that fits the patient's age and oral-preference, but the evidence is a non-randomized retrospective comparison nested in a translational ESR1/ER-expression study, not the single-arm phase 2 the trials.jsonl row described - flag the design mismatch. Benefit modest and tempered by PR-negativity. No per-term AE table in source (translational paper, no safety section); toxicities left empty accordingly.

Sun/Wang (2018) — Oncotarget

Reference:PMID 30647846 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Niraparib comparative pharmacokinetics and efficacy in BRCA-wild-type ovarian PDX
- **Indication / model:** BRCA-wild-type ovarian cancer PDX (OVC134) in female BALB/c nude mice; >30 ovarian PDX/CDX panel spanning HRD-positive and HRD-negative
- **Design:** Niraparib partitions into tumor at 3.3x plasma concentration versus 0.6x for olaparib, giving roughly 16-fold higher dose-normalized tumor exposure. The extra drug in the tumor compartment lets niraparib trap PARP and kill cells that retain functional BRCA, where olaparib fails to reach a cytotoxic threshold.
- **n:** single PDX efficacy arm plus >30-model panel; group sizes per arm not detailed in abstract
- **Effect size:** moderate — In the BRCA-wild-type OVC134 model niraparib cut tumor growth by 58% while olaparib barely moved it. Across the >30-model ovarian panel, regressions appeared in BRCA-mutant and HRD-positive BRCA-wild-type tumors, with some growth inhibition even in Myriad-defined HRD-negative models.
- **Variance:** vs olaparib head-to-head; vehicle; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Subcutaneous PDX, not orthotopic. Directly on point for this BRCA-wild-type, genomically-unstable tumor: it is the preclinical pharmacology argument for niraparib (PRIMA, all-comers) over BRCA-gated olaparib here.

Lin/Ratner (2018) — PLoS One

Reference:PMID 30444904 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Triapine plus olaparib plus cediranib in BRCA-wild-type, PARP-inhibitor-resistant ovarian cancer
- **Indication / model:** BRCA-wild-type, PARP-inhibitor-resistant EOC: SKOV3 and OVCAR3 subcutaneous in nude mice; PEO4ip intraperitoneal in SCID-Beige mice
- **Design:** Triapine inhibits ribonucleotide reductase and adds replication stress; cediranib suppresses HR via hypoxia and PDGFR blockade; olaparib traps PARP. Stacking the three converges on the same DNA-repair vulnerability in tumors that are genetically HR-proficient and have escaped single-agent PARP inhibition.
- **n:** per-arm group sizes not stated in abstract; survival cohorts
- **Effect size:** moderate — In the intraperitoneal PEO4ip model the triple combination extended median survival to 89 days versus 62 in controls and beat every lower-order arm. Supports layering anti-angiogenic and replication-stress agents onto a PARP inhibitor in BRCA-wild-type, PARP-resistant disease.
- **Variance:** vs vehicle and lower-order single/double agent arms; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Triapine is not on this patient's option list, so the directly transferable insight is the cediranib-plus-olaparib backbone in BRCA-wild-type disease. PEO4ip carries a reverted BRCA2, a resistance

model rather than a naive maintenance setting.

Moore/DiSilvestro (2018) — N Engl J Med

Reference: [PMID 30345884](#) · Type: trial · Inclusion: included

- **Intervention:** olaparib monotherapy maintenance
- **Indication / model:** newly-diagnosed advanced ovarian cancer with a germline or somatic BRCA1/2 mutation (1L); germline or somatic BRCA1/2 pathogenic mutation REQUIRED
- **Design:** randomized double-blind placebo-controlled, 1L maintenance
- **n:** 391
- **Effect size:** 0.3 — progression-free survival
- **Variance:** 95% CI 0.23–0.41; $p < 0.001$
- **Case fit:** none
- **Notes:** (toxicity table not extracted)

—/— (2018)

Reference: — · Type: trial · Inclusion: included

- **Intervention:** MAPK-pathway-directed investigational agent
- **Indication / model:** advanced solid tumors (genotype-matched basket including MAP2K4/MAPK-pathway) (2L+); MAP2K4 deletion / MAPK-pathway alteration
- **Design:** phase 1 genotype-matched basket
- **Effect size:** — — safety / ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2017)

Reference: — · Type: trial · Inclusion: included

- **Intervention:** MAPK-pathway-directed investigational agent
- **Indication / model:** advanced solid tumors (genotype-matched basket including MAP2K4/MAPK-pathway) (2L+); MAP2K4 deletion / MAPK-pathway alteration
- **Design:** phase 1 genotype-matched basket
- **Effect size:** — — safety / ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Wang/Cheng (2016) — Oncotarget

Reference: [PMID 26909613](#) · Type: preclinical · Inclusion: included

- **Intervention:** PI3K inhibition (BKM120) plus olaparib in PIK3CA-altered ovarian cancer
- **Indication / model:** PIK3CA-altered ovarian lines (SKOV3, HEYA8, IGROV1, EFO27); intraperitoneal SKOV3-luciferase dissemination xenograft in NOD/SCID mice
- **Design:** Pan-class-I PI3K inhibition with BKM120 lowers BRCA1/2 expression in responsive PIK3CA-altered

ovarian cells, manufacturing a homologous-recombination defect. That induced HRD then makes otherwise HR-proficient cells vulnerable to PARP trapping.

- **n:** multi-line panel; IP xenograft cohort (per-arm sizes not stated in abstract)
- **Effect size:** moderate — BKM120 plus olaparib was synergistic, raising DNA damage and apoptosis and shrinking tumor burden in the intraperitoneal model; cells that dropped BRCA1/2 most on PI3K inhibition responded best. Investigational rationale connecting the patient's PIK3CA amplification to the PARP/HRD axis, distinct from the foreclosed BRCA-mutation route.
- **Variance:** vs vehicle; single-agent BKM120; single-agent olaparib; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Models carry PIK3CA point mutations; the patient has PIK3CA amplification, so the dependency may differ in degree. Strictly investigational, with no approved PI3K-directed therapy in HGSC; surfaced as a mechanism-level link, not a near-term option.

Hirakawa/Mizunuma (2014) — J Ovarian Res

Reference: [PMID 24410765](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Aromatase inhibition in ER-alpha-positive ovarian cancer xenografts
- **Indication / model:** Ovariectomized nude mice with peritoneal carcinomatosis: OVCAR-3 (strong ER-alpha) versus DISS (weak ER-alpha)
- **Design:** Aromatase inhibition lowers local estrogen and blunts ER-alpha signaling in ER-positive ovarian tumors. Downstream it reduces VEGF and tumor microvessel density, tying the endocrine effect to an anti-angiogenic, anti-ascites readout.
- **n:** survival cohorts per line (sizes not stated in abstract)
- **Effect size:** moderate — Letrozole prolonged survival in the strongly ER-alpha-positive OVCAR-3 model but did nothing in the weakly positive DISS model, and lowered tumor VEGF and microvessel density. The ER-magnitude dependence fits this patient's ~80% ER-positive but PR-negative profile as a real-but-modest endocrine signal.
- **Variance:** vs untreated tumor-bearing mice; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Benefit tracked ER-alpha intensity, and this tumor is ER-positive but PR-negative, predicting a tempered response. No HGSC-specific orthotopic arm; OVCAR-3 is a serous-type line.

Murai/Pommier (2012) — Cancer Res

Reference: [PMID 23066039](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Differential PARP-trapping potency across PARP inhibitors
- **Indication / model:** DT40 isogenic chicken B-cell panel and human cancer cell lines with defined DNA-repair backgrounds; cell-free PARP-DNA retention assays
- **Design:** PARP inhibitors differ in how strongly they trap PARP-DNA complexes, and trapping (not just catalytic inhibition) drives cytotoxicity. Niraparib and olaparib are far stronger trappers than veliparib, with trapping potency tracking cell kill more closely than enzymatic IC50.
- **n:** isogenic cell-line panel; biochemical replicates
- **Effect size:** strong — Niraparib ranked among the most potent PARP-trapping agents tested, well above veliparib. The trapping-versus-catalytic distinction explains why a strong trapper can damage genomically

unstable, BRCA-proficient cells that survive a weak catalytic-only inhibitor.

- **Variance:** vs PARP-null and repair-proficient isogenic lines; veliparib/olaparib comparators; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Mechanistic cell-line work, not an ovarian in-vivo efficacy study. Foundational reference for why niraparib's trapping potency matters in a BRCA-wild-type setting; the later DT40 follow-up (Murai 2014) refines the ranking.

Mabuchi/Kimura (2012) — Br J Cancer

Reference:doi:10.1038/bjc.2012.261 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Bevacizumab maintenance after chemotherapy in ovarian carcinoma xenografts
- **Indication / model:** Human epithelial ovarian carcinoma xenografts (including HOC22) in mice; chemotherapy plus bevacizumab in concurrent, maintenance, and sequential schedules
- **Design:** Adding bevacizumab to platinum-taxane chemotherapy suppresses VEGF-driven angiogenesis during and after cytotoxic treatment. Continuing it as maintenance keeps the residual tumor vasculature suppressed once chemotherapy stops.
- **n:** multiple xenograft arms (per-arm group sizes not stated in abstract)
- **Effect size:** moderate — Bevacizumab with cisplatin and paclitaxel delayed progression, and continuing it as maintenance after chemotherapy gave the longest survival, with complete responses in the HOC22 model. Preclinical mirror of the GOG-218/ICON7 maintenance schedule this patient is on.
- **Variance:** vs cisplatin plus paclitaxel without bevacizumab; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** PMID not independently confirmed in this run; DOI verified (10.1038/bjc.2012.261). Schedule-comparison xenograft work; the maintenance arm is the load-bearing result for the patient's current bevacizumab continuation.

Burger/Liang (2011) — N Engl J Med

Reference:PMID 22204724 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Bevacizumab (with maintenance)
- **Indication / model:** newly diagnosed stage III-IV epithelial ovarian / primary peritoneal cancer (1L); All-comers, no biomarker selection. Matches the VEGF/angiogenesis axis the patient is already being treated on.
- **Design:** randomized double-blind placebo-controlled three-arm phase 3 (GOG-218)
- **n:** 1873
- **Effect size:** 0.717 HR — median PFS (bevacizumab-throughout)
- **Variance:** 95% CI 0.625–0.824; $p < 0.001$
- **Toxicities:** hypertension $G \geq 2$ (22.9%); GI perforation/wall disruption (2.6%)
- **Case fit:** strong
- **Notes:** Anchors the bevacizumab maintenance axis the patient is already on. PFS benefit only; OS was not improved in the overall population. VTE/bleeding caution is the operative concern here, not the patient's named neuropathy or myelosuppression vetoes.

Perren/Oza (2011) — N Engl J Med

Reference:PMID 22204725 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Bevacizumab (with maintenance)
- **Indication / model:** newly diagnosed epithelial ovarian / primary peritoneal cancer (1L); All-comers, no biomarker gate. OS benefit concentrated in the high-risk subset (stage IV or suboptimally debulked); this stage IIIc R0 patient sits adjacent to that subset.
- **Design:** randomized open-label two-arm phase 3 (ICON7)
- **n:** 1528
- **Effect size:** 0.81 HR — median PFS
- **Variance:** 95% CI 0.7–0.94; p=0.004
- **Toxicities:** hypertension G $\geq$ 2 (18%)
- **Case fit:** strong
- **Notes:** Reinforces the bevacizumab axis; largest benefit in the high-risk subgroup. Open-label design so RoB is some-concern. OS gain limited to the high-risk subset.

Anwer/Fewell (2010) — Gene Ther

Reference:PMID 20033066 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** IL-12 plasmid lipopolymer (GEN-1 / EGEN-001) intraperitoneal immunogene therapy: ID8 preclinical basis
- **Indication / model:** Murine ID8 intraperitoneal ovarian tumor model dosed with mouse IL-12 plasmid formulated in PEG-PEI-cholesterol lipopolymer (pmIL-12/PPC); plus first-in-human phase I
- **Design:** Local intraperitoneal delivery of an IL-12-encoding plasmid produces sustained, compartment-restricted IL-12 and IFN-gamma rather than a short systemic protein bolus. That shifts the peritoneal immune microenvironment toward a Th1 program and lowers ascitic VEGF.
- **n:** tumor-bearing mouse cohorts plus 13-patient phase I
- **Effect size:** moderate — In ID8 mice, pmIL-12/PPC raised peritoneal IL-12 and IFN-gamma, drove a Th1 gene-expression shift, dropped ascitic VEGF, and improved survival alone and with chemotherapy. Mechanistic foundation for the GEN-1/IMNN-001 program this patient received, working through the TME rather than the foreclosed MSI/TMB checkpoint axis.
- **Variance:** vs untreated/empty-vector tumor-bearing mice; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** The ID8 preclinical package is reported alongside the phase I in this and companion Immunon/EGEN sources rather than as a single standalone mouse paper; treat as the preclinical-plus-early-clinical anchor. Syngeneic ID8 immunogenicity may exceed human HGSC, and MSS/TMB-low status argues against single-agent checkpoint blockade, not against this IL-12 route.

Mesiano/Jaffe (1998) — Am J Pathol

Reference:PMID 9777956 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** VEGF immunoneutralization controls ovarian tumor growth and ascites
- **Indication / model:** Human SKOV-3 ovarian carcinoma, subcutaneous and intraperitoneal, in immunodeficient mice; treated with anti-VEGF antibody A4.6.1 (the murine antecedent of bevacizumab)
- **Design:** VEGF supports ovarian tumor growth by driving neovascularization and raises vascular permeability to produce malignant ascites. Blocking VEGF starves the angiogenic supply and shuts down the permeability that

fills the peritoneal cavity.

- **n:** xenograft cohorts (group sizes per arm not specified in abstract)
- **Effect size:** strong — A4.6.1 significantly slowed subcutaneous SKOV-3 growth, partly slowed intraperitoneal growth, and abolished ascites formation; tumor regrew once antibody was withdrawn. This is the founding preclinical rationale for the VEGF-maintenance axis the patient is already on with bevacizumab.
- **Variance:** vs untreated tumor-bearing mice; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** A4.6.1 is the precursor antibody clone, not clinical bevacizumab, and growth resumed off-therapy, mirroring the maintenance-not-cure clinical pattern. Older study; the ascites-control finding remains the cleanest mechanistic anchor for anti-VEGF here.